

Module 08: Planning in Robotics

Learning Objectives

1. Define **Planning** within the context of Robotics.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Robotics, we see Planning manifest in: * **Specific Example 1:** A desert-ant-inspired robot plans return-to-nest trajectories using a path integration mechanism (analogous to the ant's

celestial compass and step counter) that maintains a running estimate of the home vector; when foraging is complete, the robot evaluates candidate return paths by predicting sensory consequences -- a direct homeward path minimizes expected free energy when the home-vector estimate is confident, but when uncertainty is high (long foraging trip, accumulated drift), the planner selects landmark-following paths that trade directness for information gain, just as real desert ants switch from path integration to visual piloting. * **Specific Example 2:** A corvid-inspired robot planner demonstrates prospective cognition by simulating future tool-use sequences before acting -- when faced with a food item inside a tube, the robot's generative model evaluates candidate action sequences (push with stick A, then hook with stick B) by mentally simulating each step's expected sensory outcomes, selecting the multi-step plan that minimizes expected free energy across the full temporal horizon rather than greedily choosing the immediately most rewarding action, mirroring the hierarchical planning observed in New Caledonian crows.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.